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ARTICLE

Identification and Optimization of Carbon Radicals on Hydrated Graphene Oxide for Ubiquitous Antibacterial Coatings

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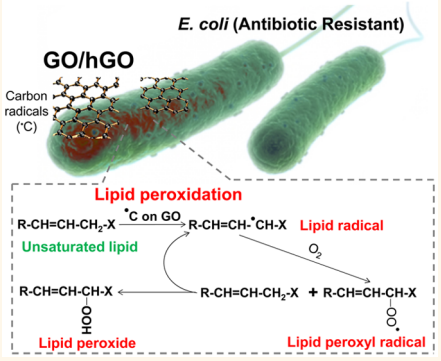
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Supporting Information

ABSTRACT: **While the antibacterial properties of graphene oxide (GO) have been demonstrated across a spectrum of bacteria, the critical role of functional groups is unclear. To address this important issue, we utilized reduction and hydration methods to establish a GO library with different oxidation, hydroxyl, and carbon radical (•C) levels that can be used to study the impact on antibacterial activity. Using antibiotic-resistant bacteria as a test platform, we found that the •C density is most proximately associated with bacterial killing. Accordingly, hydrated GO (hGO), with the highest •C density, had the strongest antibacterial effects through membrane binding and induction of lipid peroxidation. To explore its potential applications, we demonstrated that coating of catheter and glass surfaces with hGO is capable of killing drug-resistant bacteria. In summary, •C is the principle surface moiety that can be utilized for clinical applications of GO-based antibacterial coatings.**



KEYWORDS: *graphene oxide, super bugs, antibacterial coating, antibiotic resistance, functionalization*

Chemically modified graphene has been widely studied for various applications, such as polymer composites, energy-related materials, and catalysis.[1,2] Of particular interest is graphene oxide (GO), an oxygenated form of graphene decorated with abundant functional groups. GO is widely used due to its wide availability, facile synthesis, and outstanding electronic, optical, and chemical properties.[3−5] A major use for this material is in biomedical applications for drug delivery, biosensors, and tissue engineering due to its high dispersibility, two-dimensional (2D) planar structure, large surface area, and surface functionalities.[6] In addition, GO has broad-based antibacterial effects that require detailed structure−activity relationships (SARs) to be established.[7−9]

There are two major schools of thought on the mechanism of GO-induced bactericidal effects. One is relying on the physical interaction between the unique GO structure and bacterial membranes, including direct physical puncturing of the bacteria membrane,[8] or destructive extraction of lipid molecules from the bacterial membrane.[9] However, this hypothesis is mostly premised on computer simulation, and there is a paucity of experimental evidence regarding the details of damage to the bacterial surface. Another mechanistic explanation, with supportive experimental data, is oxidative damage to the bacterial membrane by the generation of reactive oxygen species and charge transfer.[10−12] While surface functional groups are assumed to play a critical role in mediating GO oxidative damage, the complex chemistry related to this material has made it difficult to discern the exact surface functional groups that may be involved in this outcome.

As-prepared GO has different oxidation levels and surface functional groups such as the presence of epoxy (−COC−), hydroxyl (−OH), and carboxyl (−COOH) moieties at

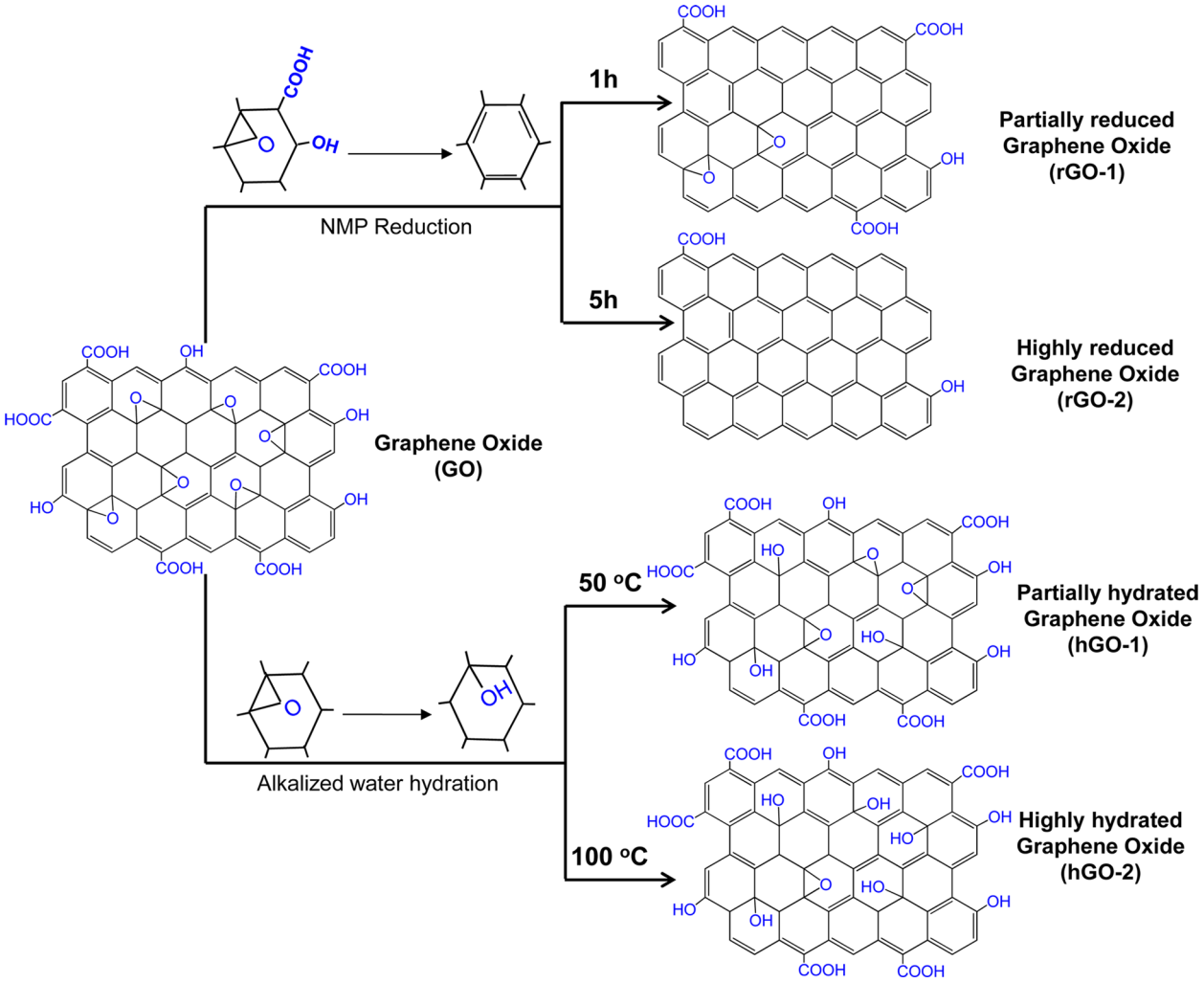
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Scheme 1. Synthesis of Reduced and Hydrated GOs[a]

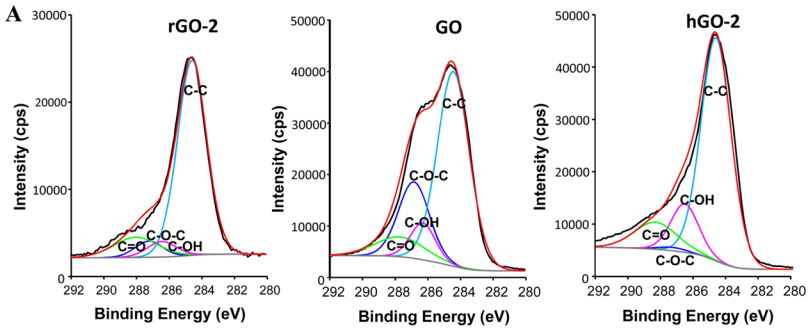


[a]GO was synthesized by a modified Hummers method. rGO-1 and rGO-2 were synthesized by solvothermal reduction of GO in *N*-methyl-2-pyrrolidinone at 150 °C for 1 or 5 h, respectively. To prepare hGO-1 and hGO-2, GO was hydrated in aqueous alkalized solution at 50 or 100 °C for 24 h. Reaction of the epoxy groups with nucleophiles leads to the opening of these rings and the generation of hydroxyl groups (as well as •C radicals shown in Figure 1B).

different densities and combinations.[13] In addition, isolated electrons in the carbon p orbitals are often conjugated by π bonding, which could form carbon radicals (•C) at discrete sites on the material surface.[14] Although attempts have been made to explore the role of oxidation level,[13,15] lateral flake size,[16] or catalytic capability[11] on bacterial killing, results have been inconclusive and even contradictory. One reason is the interlinked complexity of the functional groups, such that a change in one surface group will also affect others, often in a nonpredictable fashion. To overcome this challenge, a new approach is required to change the surface functionalities in a more systematic fashion to elucidate SARs. For instance, it has been shown that solvothermal reduction could quantitatively adjust GO oxidation levels,[17] while hydrolysis by alkalized aqueous solvents could open epoxy rings and quantitatively adjust the hydroxyl density.[18] While it is also possible that the •C levels could also be impacted by these processes,[19] this aspect has not been studied. These evolving concepts prompted us to hypothesize that quantitative adjustment of GO surface functionalities may allow us to establish SARs for bacterial killing by the various surface groups.

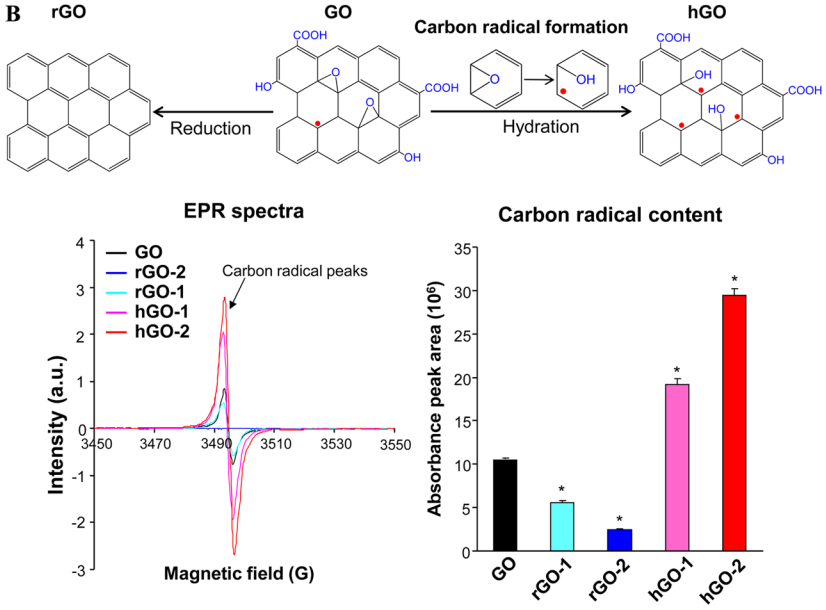
Herein, using reduction and hydration methods, we successfully synthesized a library of GO with different surface functionalities and compared their antibacterial effects in *Escherichia coli* (*E. coli*), which is a popularly used Gram-negative bacterium to explore new antibacterial agents and study antibiotic resistance.[9,12] Through detailed characterization and toxicity testing, we identified •C as the functionality that is best associated with the antibacterial effects. The hydration method could quantitatively modulate the •C density, allowing hydrated GO (hGO) with the highest radical levels to exert the most effective antibacterial effects. To explore its potential applications, we coated the surfaces of glass slides and silicone catheters with hGO for proof-of-concept experimentation. Immobilized hGO provided highly effective bactericidal effects in antibiotic-resistant (AR) *E. coli*, commensurate with the coating density. These results demonstrate the important role of •C and potential use of GO coatings on medical devices to combat antibiotic resistance.

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Atomic % concentration of oxidized groups on GO surface by XPS

	rGO-2	GO	hGO-2
C=O	8.3	9.4	12.2
C-O-C	4.1	20.5	1.1
C-OH	4.1	6.8	14.5
Total oxygen	16.5	36.7	27.8



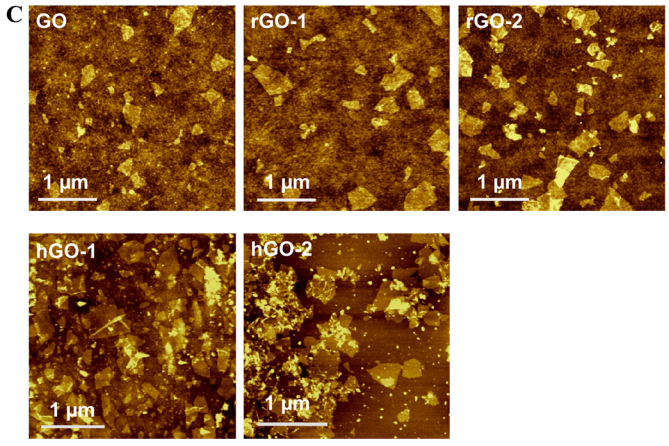


Figure 1. Characterization of GO library materials by EPR, XPS, and AFM. (A) XPS spectra of oxidized groups on the GO surface. XPS was performed by the stepwise (50 meV) acquisition of high-resolution spectra of the C 1s region. (B) Detection of •C on the GO surface by EPR. EPR was used to assess the •C density on GO surface by testing 5 mg of each of the dried GO samples by an X-band Bruker ELEXYS 580 spectrometer with a g value of 2.0029. (C) Visualizing the morphology of GO samples by AFM. AFM images were obtained by placing a drop of the GO solution (10 μg/mL) on Si wafers that were pretreated with a 2.5 mM (3-aminopropyl)triethoxysilane aqueous solution. After being washed with water and dried with N_2, AFM images were obtained in an Asylum Research Cypher ES AFM in tapping mode with conical probes.

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Table 1. Hydrodynamic Size and ζ-Potential of GO Samples in Aqueous Solution

nanoparticles		GO	rGO-1	rGO-2	hGO-1	hGO-2
primary size (nm)		148	152	148	157	138
hydrodynamic size (nm)	water	334.1 ± 3.1	378.1 ± 3.9	1349.2 ± 8.5	307.5 ± 6.7	329.8 ± 7.0
	LB	655.4 ± 95.4	615.9 ± 54.9	474.9 ± 14.9	412.1 ± 4.5	533.8 ± 26.6
	Lacto	443.6 ± 4.6	654.9 ± 44.3	509.6 ± 6.7	367.2 ± 8.4	409.5 ± 48.3
ζ-potential (mV)	water	−51.7 ± 0.9	−36.3 ± 1.4	−24.4 ± 0.7	−50.6 ± 0.8	−49.4 ± 0.9
	LB	−18.53 ± 0.6	−8.7 ± 2.7	−8.2 ± 2.2	−16.7 ± 0.8	−14.7 ± 2.5
	Lacto	−7.5 ± 0.5	−5.5 ± 1.2	−6.5 ± 3.6	−7.9 ± 3.9	−6.4 ± 2.2

RESULTS

Establishing a Well-Characterized GO Material Library with Varying Surface Functional Groups. Recent developments in the functionalization of the GO surface present the opportunity for chemical tuning of these groups for the purpose of directed work functions. Oxidation levels can be modified through the use of reduction processes based on solvothermal or chemical methods for the production of reduced graphene oxide (rGO), which is chemically similar to graphene, albeit not identical (Scheme 1). During reduction, the variety and evolution of oxygenated species on the GO surface can be tracked by methods such as X-ray photoelectron spectroscopy (XPS),[20,21] Raman spectroscopy,[21] and electron paramagnetic resonance (EPR).[14] It is also possible to use hydration chemistry for opening epoxy rings through hydrolysis, with the ability to change the density of surface hydroxyl groups; this can be achieved by heating GO in an alkaline environment (Scheme 1).

In order to assess the SARs most directly involved in bacterial killing, we established a GO material library by using reduction and hydration methods to change surface functionalities. Pristine GO prepared by the classic Hummers method was used as the base material for preparing the library. The total level of oxidized groups on the GO surface was reduced by solvothermal reduction in *N*-methyl-2-pyrrolidinone (NMP), while heating in alkaline water leads to an increased hydroxyl density by hydrolyzing the epoxy surface groups (Scheme 1). Through control of the reaction conditions (time or temperature), we prepared a series of GO samples with quantifiable differences in their oxidation states and surface functional groups. From these materials, we selected two reduced GO (rGO-1 and rGO-2) and two hydrated GO (hGO-1 and hGO-2) samples for comparative study *versus* pristine GO. XPS was used to determine the levels of oxidized functional groups on the GO surface. The C 1s XPS spectra of the rGO-2, GO, and hGO-2 sheets (Figure 1A) show C═C bonds at 284.6 eV, C−OH at 286.5 eV, C−O−C at 287.3 eV, and C═O at 288.0 eV, as previously described.[22,23] Analysis of the XPS peaks allowed us to estimate the atomic % of −COOH, −COC−, and −OH groups on the GO surface. This analysis showed that the NMP reduction could significantly reduce the peak intensities of the total oxygen content as well as the % of each oxidized group on the material surface. Different from the reduction process, hydrolysis mainly reduces the C−O−C content on the GO surface. The intensity of the C−O−C peak decreased to 1.1%, while the C−OH peak increased from 6.8 to 14.5% for hGO-2. These changes likely result from the reaction between the epoxy groups and nucleophiles in aqueous solution, leading to the generation of −OH groups. The total oxygen levels remained stable during hydration, likely due to the fact that opening of the epoxy rings will not change the oxygen levels.

Again, we know that the isolated electrons in the p orbitals of the carbon atom are usually conjugated to the GO backbone by π bonding, which can result in the formation of •C.[14] Thus, we also characterized the •C density on the GO surface through an EPR method. Interestingly, these measurements demonstrated that, in addition to the changes in oxygen levels and hydroxyl groups as a result of reduction and hydration chemistry, there is a dramatic change in the density of •C. As shown in Figure 1B, all GO samples showed a single resonance peak of π-conjugated •C with g = 2.0029, which is consistent with a previous report.[14] In addition, we ruled out the possibility that the EPR signals on GO samples resulting from the residual oxygen radicals (such as •OH and $\bullet O_2^-$) by the 3′-(*p*-aminophenyl)fluorescein (APF) and 3-bis(2-methoxy-4-nitro-5-sulfophehyl)-2*H*-tetrazolium-5-carboxanilide) (XTT) assays (Figure S1). hGO samples showed EPR peaks significantly higher than those of pristine GO (with hGO-2 being the highest), while rGO samples showed a radical density lower than that of GO. rGO-2 exhibited the weakest EPR signal. These changes suggest that hydrolysis of −COC− groups is accompanied by the generation of •C, which stably exist on the planar GO surface as π-conjugated •C.[14] In addition, we demonstrated that the EPR signals used to establish a role for •C show the utility of reduction and hydration chemistry to quantitatively change the surface functionalities of GO sheets.

We also thoroughly characterized other physicochemical properties of the GO library materials, including primary size, shape, hydrodynamic size, and surface charge in aqueous solution. Atomic force microscopy (AFM) showed that all of the GO materials were irregularly shaped nanosheets with a lateral size distribution of 50−300 nm (Figure 1C). Raman spectra demonstrated that rGO, GO, and hGO samples preserved the graphene structure, with maintenance of the signature G and D bands (Figure S2); this suggests that there are no major structural changes during the catalytic modification of the GO surface. We also assessed the hydrodynamic size distribution and ζ-potential of the GO materials in different aqueous media. As shown in Table 1, all of the GO nanosheets showed approximately the same hydrodynamic size distribution (400−700 nm) and negative ζ-potential values (5 to −19 mV) in the presence of LB or Lactobacilli MRS (Lacto) broths (Table 1).

Carbon Radical Density Is the Determining Factor in the Antibacterial Effects of GO. To study the killing effects of pristine GO on bacteria, we selected wild-type and AR *E. coli* as well as AR *Lactobacillus crispatus* (*L. crispatus*) as representative Gram-negative and Gram-positive bacterial strains, respectively. The AR strains were included in light of the public health relevance of AR bacteria. The chosen *E.*

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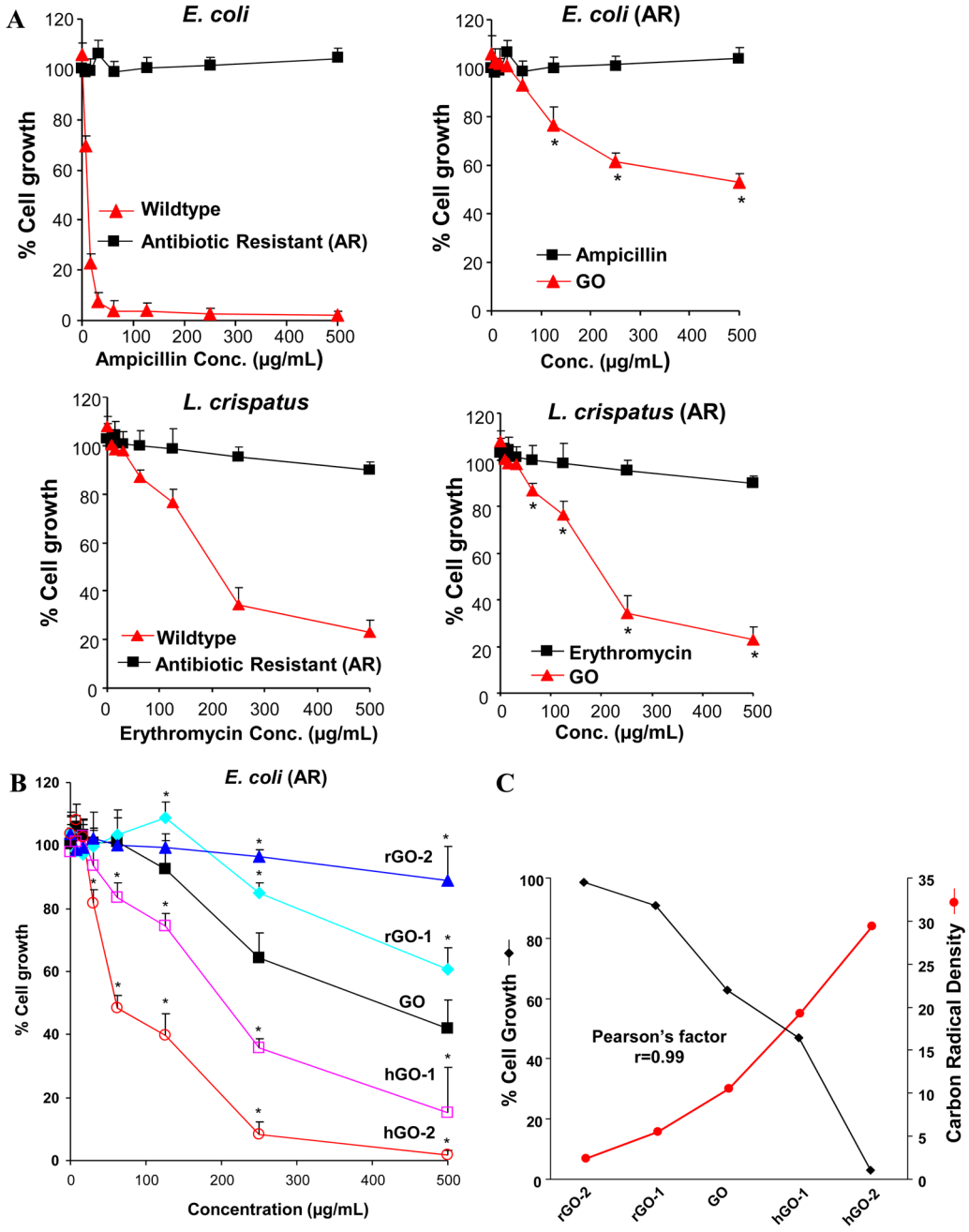


Figure 2. Bactericidal effects of the GO library. (A) Comparison of bacterial killing by GO and antibiotics in sensitive (wild-type) and AR bacteria. The resistance of bacteria was demonstrated by comparing the killing effect of antibiotics (ampicillin or erythromycin) in sensitive and AR *E. coli* (top left panel) and *L. crispatus* (bottom left panel) after incubation at 37 °C for 24 h; the bactericidal effects of GO were compared with those of ampicillin in AR *E. coli* (top right panel) and erythromycin in AR *L. crispatus* (bottom right panel). Cell growth was evaluated by measuring bacterial OD at 600 nm; *$p < 0.05$ compared to ampicillin or erythromycin. (B) Bacterial killing effects of GO samples in resistant *E. coli*. To determine the bacterial killing by GO samples, AR *E. coli* were exposed to 8−500 μg/mL rGO-2, rGO-1, GO, hGO-1, and hGO-2 at 37 °C for 24 h; *$p < 0.05$ compared to GO. (C) Calculation of the correlation coefficient between •C density and bacteria killing. Pearson's analysis was used to evaluate the correlation between the rate of cell growth and the absorbance peak area of •C on the GO surface.

coli strain is resistant to 24 different types of antibiotics, including ampicillin (Table S1). This is demonstrated by the fact that ampicillin could induce effective killing with an IC_{50} of 10 μg/mL in wild-type *E. coli*, while the resistant strain showed no growth inhibition, even at a dose as high as 500 μg/mL (Figure 2A). In contrast, GO showed moderate antibacterial effects in wild-type and AR *E. coli*, yielding an IC_{50} of 500 μg/mL. Similar results were found in the AR *L. crispatus* strain (77% killing at 500 μg/mL), as well as in silver-resistant *E. coli* (Figure S3).

Since GO only exerts partial antibacterial effects, it would be desirable to improve the killing efficiency. This prompted us to look at the antibacterial effects of the GO library. We found that hGO sheets have significantly enhanced antibacterial effects compared to pristine GO, with hGO-2 achieving almost 100% killing at 500 μg/mL (Figure 2B).

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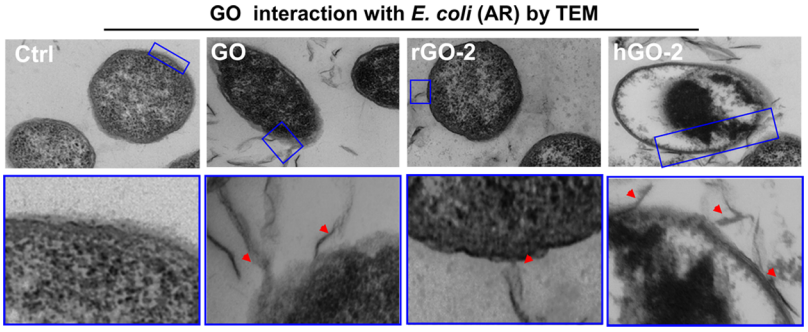


Figure 3. TEM images showing the interaction between GO samples and AR *E. coli*. Cells were treated with 125 μg/mL of each of the GO samples for 24 h, washed, embedded in resin, and negatively stained before TEM imaging. Arrows show the rGO-2, GO, or hGO-2 nanosheets interacting with bacteria membrane.

In contrast, rGO-1 and rGO-2 showed decreased bactericidal effects compared to the pristine material. Comparison of the antibacterial efficiency with the physicochemical characteristics of the library materials (including oxidized chemical content, •C density, sheet size, and ζ-potential) showed the strongest correlation (r = 0.99) to •C density (Figure 2C). However, as expected, the correlation coefficient to the hydroxyl content was also strong (r = 0.92) in light of the intimate relationship with the •C content. These results demonstrate the important role of •C in GO-induced antibacterial effects in AR bacterial strains.

GO-Induced Bactericidal Effects Involve Membrane Binding and Lipid Peroxidation. It has been shown that the hydrophobic planar properties and hydrophilic edges of GO sheets allow membrane insertion, which could be related to their bactericidal effects.[24] In order to obtain ultrastructural resolution of the interaction of our library materials with the AR *E. coli* strain, transmission electron microscopy (TEM) analysis was performed after bacterial exposure in culture medium for 24 h. Due to the low electron densities of GO, the suspended GO sheets could only be visualized when vertically positioned on the grid but not when they aligned with the flat surfaces parallel to the grid (Figure S4). This is consistent with other reports.[9] Following bacterial coculture, the TEM images revealed that the GO nanosheets associate with the bacterial membrane without clear evidence of piercing or cutting of the bacterial membrane (Figure 3); this is consistent with the observations of Tu *et al.*[9] We also observed that while the pristine and reduced (rGO-2) sheets may associate with the membrane without any visible damage, AR *E. coli* treated with hGO-2 displays extensive structural damage, as manifested by membrane collapse.

The •C radicals are typically considered more active than other surface functional groups because they contain unpaired electrons that lead to pro-oxidative potential.[14] The pro-oxidative capacity of the GO library was examined by 2′,7′-dichlorodihydrofluorescein (H_2DCF), which can be converted to a fluorescent 2′,7′-dichlorofluorescein (DCF) derivative by a variety of oxidizing radicals generated by nanomaterials or small molecules.[25−27] As shown in Figure 4A, compared to pristine GO, hGO materials induced a significant increase in DCF fluorescence, while the reactivity of the rGO materials were low. Thus, the overall ranking of the abiotic oxidative potential of the GO samples was hGO-2 > hGO-1 > GO > rGO-1 > rGO-2, which correlates well with the •C densities (r > 0.97). hGO-2 was also used to evaluate the stability of the carbon radicals on the material surface, using serial EPR measurements. This demonstrated limited change in the EPR peak intensity and pro-oxidative capability for hGO-2 samples stored in deionized water at 4 °C for 1 year (Figure S5). However, the carbon radical existence on GO samples in biological media will likely be of short duration due to the presence of abundant reductive molecules.

To further demonstrate how the pro-oxidative effects under abiotic conditions may relate to membrane injury by hGO-2, a BODIPY 581/591 C11 reagent was used to study lipid peroxidation in *E. coli*. The reagent shows a shift in peak fluorescence emission from red (∼590 nm) to green (∼510 nm) during lipid peroxidation, as demonstrated by using cumene hydroperoxide (CH) as a positive control (Figure 4B). hGO-2 induced prominent lipid peroxidation compared to pristine GO, while rGO-2 had minimal effects. These results were also quantitatively expressed by studying lipid peroxidation with flow cytometry, which showed that pristine GO induced lipid peroxidation in 12% of bacterial cells, while the comparable values for hGO-2 and rGO-2 were 21 and 5%, respectively (Figure 4C). Since lipid peroxidation could destabilize the bacterial membrane, we studied β-galactosidase release from AR *E. coli* after GO treatment. We found that hGO-2 can induce 22% β-galactosidase release from membrane-damaged bacteria (Figure 4D), which is significantly higher than GO (8%) or rGO (3%). In summary, these data demonstrate that the bactericidal effects of GO are dependent on interactions with the bacterial membrane, where surface reactive groups such as the •C could lead to membrane damage and cell death as a result of lipid peroxidation. Figure 4E shows the proposed mechanism for the induction of lipid peroxidation as a result of the •C on the GO surface interacting with unsaturated lipids in the bacterial membrane. We propose the involvement of three steps, namely, (i) electron transfer from the •C to one of the C atoms adjacent to double bonds in the lipid; (ii) electron transfer from this functionalized C atom to bystander molecular dioxygen (O_2), leading to the formation of a lipid peroxide radical, containing the attached −O−O group; and (iii) generation of a lipid peroxide from the lipid peroxide radical.

Highly Efficient Killing of AR Bacteria by hGO-2 Films. Because GO is often incorporated into composites (*e.g.*, bandages and biofilms) for antibacterial activity, we were interested to see whether immobilization of hGO to material surfaces could attain significant bactericidal effects. We performed proof-of-principle testing by depositing hGO-2 thin films on a glass substrate or covalently bonding the hGO

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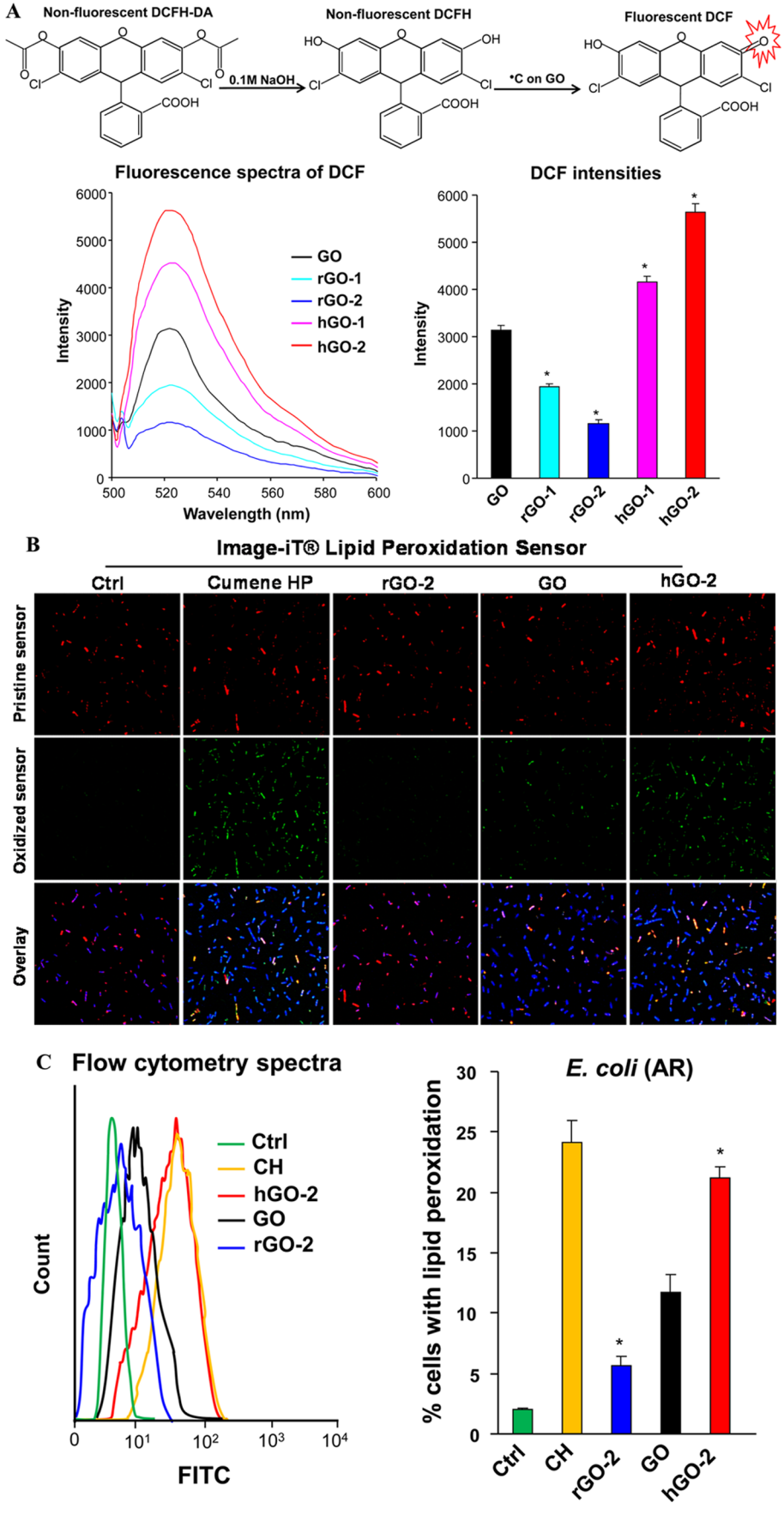


Figure 4. continued

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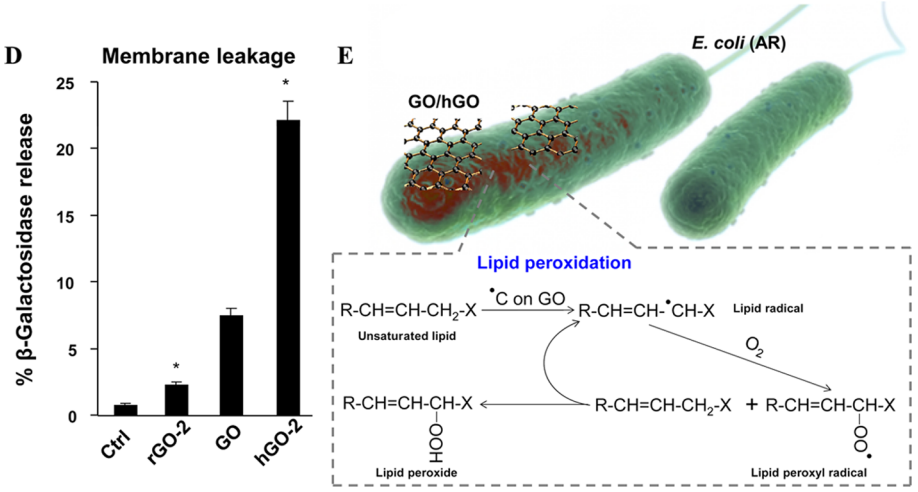


Figure 4. Mechanism of GO bacterial killing involves membrane association, lipid peroxidation and membrane damage. (A) Determination of the pro-oxidative potential of GO samples by an abiotic DCF assay; to assess the oxidation potential of GO samples, 95 μL aliquots of 25 μg/mL DCF were added into each well of a 96-multiwell black-bottom plate and mixed with 5 μL of GO suspensions at 5 mg/mL, followed by 2 h incubation. DCF fluorescence emission spectra were recorded using a SpectraMax M5 microplate reader with an excitation wavelength of 490 nm. (B) Confocal imaging of GO-induced lipid peroxidation in AR *E. coli*. To assess lipid peroxidation, AR *E. coli*, prior exposed to 250 μg/mL GO samples or 10 μM cumene hydroperoxide, was stained by using the Image-iT lipid peroxidation sensor kit and Hoechst 33342 dye before performance of confocal microscopy. The microscopy was carried out using a Texas Red filter to visualize the reduced dye and a fluorescein isothiocyanate (FITC) filter to visualize the oxidized dye. (C) Quantification of cells with lipid peroxidation by flow cytometry. Flow cytometry was carried out on a FACS Vantage SE flow cytometer to determine the percentage cells undergoing membrane lipid peroxidation. (D) β-Galactosidase release in GO-treated bacteria. The permeability of the cytoplasmic membrane was evaluated using a luminescent β-galactosidase substrate to measure the activity of released β-galactosidase from *E. coli* treated by 250 μg/mL GO suspensions; *$p < 0.05$ compared to GO treatment. (E) Schematic image to explain the bactericidal effect of GO including membrane association and lipid peroxidation.

to the surfaces of silicone catheters. hGO-2 nanosheets were attached to the surface of a glass substrate by spin-coating, allowing us to prepare a series of coated substrates with controllable GO film thickness (7−25 nm) and surface coverage. Four glass substrates (S-1 to S-4) coated with hGO-2 films (Figure S6A) were characterized through the use of AFM (Figures 5A and S6B) and Raman spectroscopy (Figure S7). The Raman spectra demonstrated the presence of signature D and G bands that are characteristic of GO, and the AFM images confirmed the varying thickness of these films, which varied from 7 to 25 nm. Moreover, AFM showed that hGO-2 coating exhibited interesting stacking behavior with larger sheets attaching to the substrate with their flat sides facing up, while smaller sheets were stacked on top of the larger sheets. The surface area coverage varied from 50% in S-1 to 100% in S-4.

Testing of the antibacterial effects of the coated glass surfaces was carried out by incubating S-1 through S-4 with the AR *E. coli* strain for 6 h at 37 °C. The bacterial morphology and viability were subsequently assessed by scanning electron microscopy (SEM) and propidium iodide (PI) staining, respectively. Instead of the rod-shaped appearance and intact surfaces found on non-exposed bacteria, SEM demonstrated an incremental increase in bacterial membrane damage and fragmentation during exposure to the incremental stacking density in S-1 to S-4 (Figure 5B). Assessment of the bacterial viability by PI staining and confocal microscopy showed that while there were <10% PI-positive bacteria grown on the uncoated glass surface, there was a gradual increase in the presence of dead cells with incremental levels of coating, such that 96% of bacterial cells were nonviable in the presence of S-3 and S-4. It is also noteworthy that the killing effects of the hGO-2 films were more effective than that of the hGO-2 suspensions, suggesting that a coated film format may be superior for antibacterial applications.

The next proof-of-concept testing was covalent attachment of hGO-2 to the surface of a silicone catheter, which has relevance from the perspective of bacterial contamination of indwelling clinically used devices.[28] A method was developed to prepare the substrate surface for coating, namely, surface amination by (3-aminopropyl)triethoxysilane (APTES), followed by conjugation of the amine groups with hGO-2 (Figure 6A). These coatings showed extraordinary durability and could stably attach to the catheter surface, even after sonication and washing procedures. This concept was initially tested on a silicon wafer, where the coating could be confirmed by the signature G and D bands in the Raman spectra that were obtained, as well as the presence of a GO coating layer by AFM after 10 cycles of coating (Figure S8). The procedure was then repeated on a series of silicone catheters that were coated with hGO-2 for 1, 3, 5, and 10 cycles (denoted as 1C, 3C, 5C, and 10C). Each layer of coating resulted in progressive darkening of the catheter surface. The antibacterial effects of the coated catheter surfaces were tested *in situ* by visualizing morphological changes in AR *E. coli*. As shown in Figure 6B, while the bacteria incubated with uncoated catheter surfaces exhibited rod-shaped cells with intact surfaces, holes appeared in the bacterial membrane during exposure to coated catheters. We also observed bacterial fragmentation during exposure to catheter surfaces coated 5 or 10 times. These results were confirmed in a β-galactosidase assay, which reflects progressive bacterial membrane leakage (Figure 6C). We also performed a quantitative evaluation of the bactericidal effects of the 10C hGO-2-coated catheter surface. As shown in Figure S9, after

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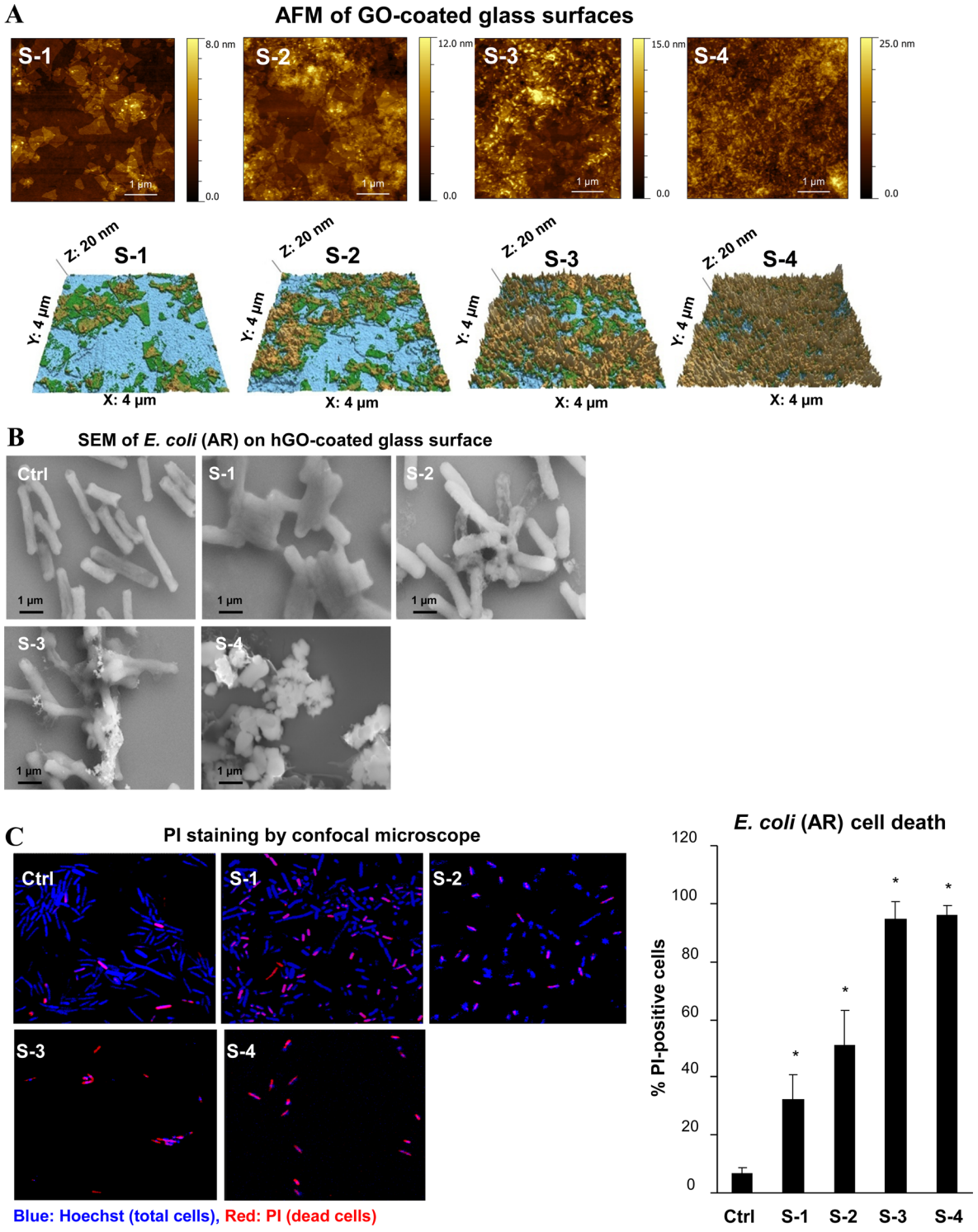


Figure 5. Inhibition of AR bacteria growth by noncovalently coated hGO-2 films on a glass substrate. (A) AFM imaging of hGO-2 coated substrates. A series of substrates (S-1, S-2, S-3, and S-4) with different GO coverage and thickness were characterized by AFM. (B) SEM imaging of changes in the bacterial morphology after incubation with the hGO-2-coated substrate. (C) Visualization and (D) quantification of bacterial death on hGO-2 films by confocal microscopy. Following 6 h incubation of AR *E. coli* with substrates S-1 to S-4, the cells were stained with PI, fixed, and washed with 70% ethanol to determine the percentages of dead cells on substrates. The morphological changes of bacteria were visualized by SEM; *$p < 0.05$ compared to control.

1, 3, and 6 h of incubation, the bacteria settling on the catheter surface were collected, serially diluted, and introduced into LB agar plates to assess the number of colony forming units (CFU). The coated catheters showed a time-dependent reduction of CFU, compared to the uncoated catheters (Figure 6D). A dramatic decrease of CFU could be seen after 1 h of exposure to a hGO-2 film, amounting to 2.2 log reduction after 6 h exposure. All considered, the above results show that hGO-2 coating of glass and catheter surfaces can be used to kill AR *E. coli*.

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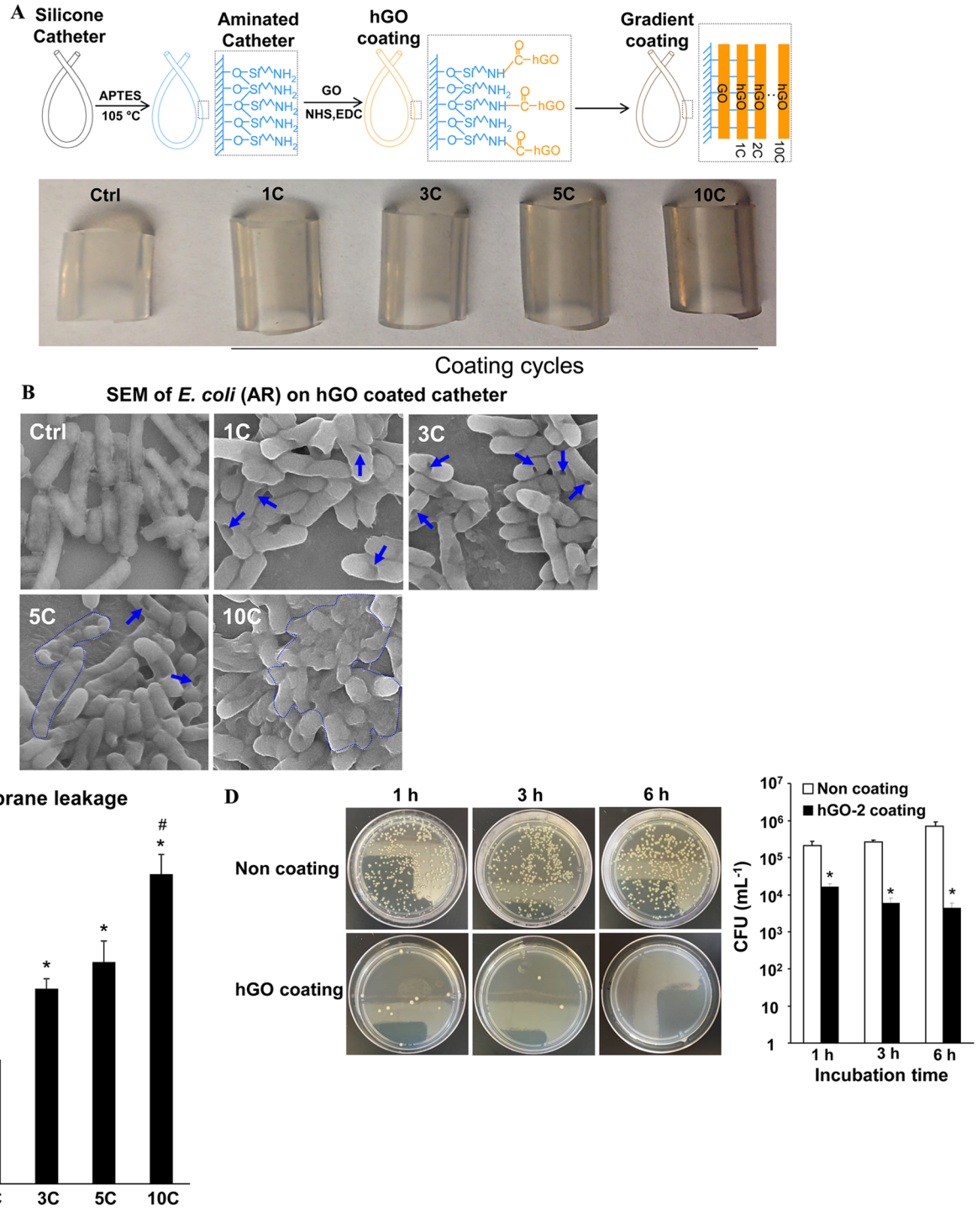


Figure 6. Inhibition of AR bacterial growth by hGO-2 covalently attached to the surface of a silicone catheter. (A) Schematic to describe hGO-2 coating of catheters as well as product images. A complete coating cycle involves the amination of catheter surface by APTES and conjugation of the hGO-2 to the amine groups. A series of hGO-2-coated catheters were prepared through multicycle coating. The catheter surfaces grow darker with incremental rounds of coating. (B) *In situ* visualization of *E. coli* bacteria embedded on coated catheters. Catheters were immersed in the bacterial suspensions to allow bacterial attachment to the catheter surface. After 6 h incubation, the catheters were treated with 2% glutaraldehyde, followed by 70% ethanol washing to remove residual salts and Au/Pt coating. The morphology of *E. coli* on catheters was observed by SEM. Arrows and the dashed-line areas show the damaged bacteria. (C) *β*-Galactosidase release from bacteria grown on coated catheters. The *β*-galactosidase release from embedded bacteria on catheter surfaces was determined after 2 h incubation. (D) Assessing the growth of bacteria retrieved from the coated catheter surfaces by CFU assay. After bacteria settled on catheter surfaces, they were incubated for 1, 3, and 6 h. The retrieved bacteria from catheter surfaces were serially diluted and spread on the agar plates for 24 h incubation. The images show the growing colonies from uncoated or hGO-2-coated surfaces at each time point at same dilutions (right panel). CFUs were calculated by desired colony numbers (20−300) at appropriate dilutions (left panel).

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DISCUSSION

In this study, we delineated the GO surface groups that are responsible for bactericidal effects and explored the potential use of GO coating on the surface of a medical device that may be contaminated by AR bacteria. We found that the •C density on the GO surface is most proximately associated with the antibacterial effects of these materials compared to other functional groups. The efficacy of bacterial killing could be enhanced by a novel hydration process of the GO surface that opens the epoxy rings to generate more surface •C groups, without impacting superoxide or hydroxyl radical generation. The high •C density enhanced the oxidative potential of hGO, which allowed these sheets to induce lipid peroxidation of the bacterial membrane with significantly higher bactericidal effects in AR strains. In contrast, surface reduction decreases •C density and antibacterial killing by GO. Coating of hGO on glass and silicone catheter surfaces allows effective killing of AR *E. coli* and may be useful for preventing biofilm formation on catheter surfaces.

The most important finding of this study is the demonstration that, among the surface reactive groups, •C radicals appear to play an important role in shaping the antibacterial activity of GO. The •C radicals are generated during breaking of the C═C bonds when graphite is treated with strong oxidants to prepare GO. We found that alkaline hydration could increase the •C density during hydrolysis of the epoxy groups on the GO surface. The formation of this radical group was reported to be an intermediary step during occupancy of the p orbitals by single electrons.[19] While isolated •C radicals are often unstable in association with a small molecule, the single carbon electrons on the GO surface are located in the π-network plane, allowing them to conjugate with the π electrons of neighboring double bonds to form π-conjugated •C,[14] which are more stable. Moreover, the mobility of •C among the conjugated C═C bonds could significantly enhance its oxidizing capability and reactivity with target molecules (*e.g.*, membrane lipids), coming into contact with the basal GO plane. This allows hGO-2, which has the highest •C density, to exert the highest bactericidal effects, in contrast to rGO that exhibits a low •C density and lack of bactericidal effects. The same characteristics are involved in the effective bactericidal effects of hGO when coated on the surfaces of catheters or a glass substrate.

Multidrug resistance in a growing number of infectious agents is considered as one of the greatest threats to human health in a recent World Health Organization report.[29] For example, fluoroquinolone-resistant *Shigella* caused more than 1 million deaths in 2012, while multi-drug-resistant *Mycobacterium tuberculosis* resulted in an estimated 170 000 deaths in the same time frame. Moreover, bacterial antibiotic resistance now impacts most pharmaceutical antibacterial agents. Among the major pathways leading to antibiotic resistance are acquired mechanisms that block antibiotic access to intracellular targets, through either the reduction of membrane permeability[30] or increased drug efflux by membrane pumps.[31] Another resistance mechanism involves the mutation of antibiotic targets in bacteria[32] or their modification[33] to prevent antibiotic binding. A third mechanism is the inactivation of antibiotics by enzymatic activities such as hydrolysis[34] or postmodification.[35] In addition, these diverse resistance mechanisms often work synergistically to establish a formidable defense against antibiotics. In contrast, GO is capable of bacterial killing by anchoring to the membrane and inducing lipid peroxidation without the necessity of cellular uptake. This circumvents traditional AR mechanisms, with the ability to overcome antibiotic and even silver resistance pathways.

To take full advantage of the antibacterial effects of GO, we postulated that it is possible to enhance its 2D material effects through modification of its surface reactivity and immobilization on the surface of a medical device that can be colonized by AR bacteria. Indeed, GO has often been integrated into products such as antibacterial paper,[36] bandages,[11] and ultrafiltration membranes.[37] Another consideration is the reported toxicity of GO to mammalian cells and animals after intravenous, intraperitoneal, subcutaneous, or intramuscular injection.[38,39] Immobilized GO films on device surfaces would minimize direct exposure of GO to mammalian cells and thus reduce the potential toxicity. Herein, we report the preparation of hGO-2 films by noncovalent and covalent attachment to glass and catheter surfaces, respectively. Since silicone catheters are widely used in medical devices and serve as the substrate for bacterial biofilms,[28] we explored stable, covalent attachment of hGO-2 that has potent antibacterial effects. hGO-2 also serves the same purpose if noncovalently attached to glass substrates. Both types of coatings are dependent on coating density and thickness for the efficacy of bacterial killing. This result could be attributed to two advantages of films over GO sheet suspensions: the first is enhanced binding to the bacterial surfaces and the second involves stronger exposure and shorter traveling distance for the reactive •C radicals on the hGO surface. In terms of the stability of spin-coating *versus* covalent attachment of hGO-2 films on a glass substrate or silicon catheter surface, respectively, we do observe some differences. While spin-coating can resist scratching or gentle washing, covalent attachment is more durable, capable of withstanding sonication for 30 min. There were also no observed differences between fresh-made and aged hGO-2 films in terms of the attachment stability over 6 months. In addition to the surface coating of medical devices, hGO-2 has the potential to be incorporated into bandages for wound healing, antibacterial additives in dental devices, or water filtration membranes to prevent biofilm formation. Obviously, further research is required on each of these fronts.

CONCLUSIONS

Our study indicates that •C plays a critical role in GO-induced bactericidal effects. Hydration of pristine GO generates a high •C density on the GO surface, which significantly enhances antibacterial effects, including in AR and silver-resistant bacteria. The mechanism of bacterial killing involves GO contact with the bacterial membrane without cellular uptake. The subsequent induction of lipid peroxidation in the bacterial membrane leads to a lethal effect. To explore the potential use of these structure−activity relationships for medical devices, hydrated GO with a high •C density was deposited onto the surfaces of glass substrates and silicone catheters, with the ability to kill AR *E. coli*. These results demonstrated the potential of GO with superior bacterial killing ability to be considered for their antibacterial effects in the clinical setting of antibiotic resistance.

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MATERIALS AND METHODS

Chemicals. Ampicillin, *N*-hydroxysuccinimide, ethyl-(dimethylaminopropyl)carbodiimide, 1-methyl-2-pyrrolidinone, erythromycin, and $AgNO_3$ were purchased from Sigma-Aldrich (St. Louis, MO, USA). The Beta-Glo assay system was purchased from Promega (Madison, WI, USA); graphite flakes were provided by Asbury Graphite Mills; Hoechst 33342, propidium iodide, and Image-iT lipid peroxidation kit were purchased from Life Technologies (Grand Island, NY, USA).

Acquisition and Synthesis of a Surface-Functionalized GO Library. *GO Production.* GO was prepared using a modified Hummers method as described previously.[40] Briefly, graphite flakes (Asbury Mills 3061 grade) were oxidized and then filtered and centrifuged to remove any residual contaminants. The oxidized graphite was then redispersed in NMP (ACS reagent grade 328634, Sigma-Aldrich) and ultrasonicated using a Fisher Scientific model 500 sonic dismembrator for 1 h at 50% power (~55 W) in an ice bath. The GO was then centrifuged using a Beckman Coulter J26-XPI at 5000 rpm for 10 min (~4620*g*). The top 80% of the supernatant was retained as the final GO sample for further processing of the reduced graphene oxide samples. The concentration of the GO solution was determined by optical absorbance at 330 nm.

Synthesis of Reduced GO. GO was dispersed in NMP by ultrasonication for 1 h at 50% power (~55 W) as described before.[41] The solution was heated to 150 °C with constant stirring in a silicone oil bath for 1 h (rGO-1) or 5 h (rGO-2).

Preparation of Hydrated GO. A 10 mL GO suspension (5 mg/mL) was diluted with 90 mL of deionized (DI) water and mixed with 80 mg of NaOH (0.02 M), using dispersal by a sonication probe (Sonics & Materials, USA) at 32 W for 10 min. The GO mixture was transferred into a round flask and refluxed at 50 or 100 °C in an oil bath with constant magnetic stirring for 24 h. HCl solutions (1 M) were used to neutralize the reaction. The mixture was centrifuged at 50 000 rpm for 30 min to collect the hydrated GO pellets. After being washed with DI water three times, the hydrated GO samples were dispersed in DI water and stored at 4 °C.

Physicochemical Characterization of GO Samples. To obtain AFM images, Si wafers were pretreated by 2.5 mM APTES aqueous solution for 30 min to functionalize the surface with a monolayer.[42] The wafers were rinsed with DI water and dried under N_2. A drop of 10 μg/mL GO solution was placed on the wafer, followed by washing twice with DI water (~5 s) and drying under N_2. Next, the GO AFM sample underwent heat treatment for 30 min at 250 °C. AFM images were obtained by an Asylum Research Cypher ES AFM. Images were taken at random locations in the sample and showed little variation. All images were obtained with the same tip and scanning conditions.

X-ray photoelectron spectroscopy (AXIS Ultra DLD, Kratos, UK) has been used to investigate the chemical state and calculate the atomic concentration of oxidized groups on GO surface with monochromatic Al Kα at 15 kV and 10 mA. For sample preparation, a suspension of GO samples was dropped on the silicon substrate and dried at room temperature. The data analysis and curve fitting were performed with the CasaXPS program (Casa Software Ltd., UK).

The molecular structure of all GO samples were characterized using Raman spectroscopy (Renishaw inVia Reflex, Wotton under Edge, UK) with a 785 nm near-infrared diode and a 50× objective lens. Spectra were obtained for 10 s exposure time with an accumulation of two scans in the wavenumber region of 500−2000 cm^{-1}.

The EPR measurements were obtained with an X-band Bruker ELEXYS 580 spectrometer. Five milligram GO nanosheets were dried through vacuum and allowed to settle to the bottom of the 2 mm i.d. quartz EPR tubes prior to data collection. The field was calibrated using a standard sample with a known *g*-factor (2,2-diphenyl-1-picrylhydrazyl). The EPR spectra were detected at room temperature with the frequency at 9.785845 GHz, center field at 3480 G, attenuator at 13.0 dB, and *g* value at 2.0029.

ζ-Potential and hydrodynamic size measurements of the GO suspensions were performed using a ZetaSizer Nano-ZS instrument (Malvern Instruments, Worcestershire WR, UK).

Assessing the Pro-oxidative Capability of GO by DCF, XTT, and APF Assays. The oxidation capacity and oxygen species of GO samples were determined as described before.[25] 2′,7′-Dichlorodihydrofluorescein diacetate (H_2DCF-DA) was used to evaluate the total oxidation capacity of GO, while the APF and XTT assays were used to determine $\bullet O_2^-$ and •OH on the GO surface, respectively. The DCF working solution was prepared by mixing 50 μg of H_2DCF-DA with 280 μL of 0.01 M NaOH. The resulting solution was incubated for 30 min and diluted with 1720 μL of a sodium phosphate buffer (25 mmol/L, pH 7.4) to form 25 μg/mL DCF solution. Ninety-five microliter aliquots of 25 μg/mL DCF, 100 μM XTT, or 10 μM APF working solutions were added into each well of a 96-multiwell black plate (Costar, Corning, NY). A 5 μL amount of 5 mg/mL nanoparticle suspension was subsequently added to each well, followed by 2 h incubation. DCF fluorescence emission spectra in the range of 500−600 nm were collected using a SpectraMax M5 microplate reader with an excitation wavelength of 490 nm. APF fluorescence emission spectra were collected at 480−600 nm with an excitation wavelength of 455 nm, while XTT absorbance spectra were recorded in the range of 410−550 nm.

Determination of Bacteria Killing by GO. Five bacterial strains were used in this study: a wide-type *E. coli* strain (ATCC 25922), an AR *E. coli* strain (ATCC BAA-2452), a silver-resistant *E. coli* strain (kindly donated by Dr. Susanne Sütterlin from the Department of Clinical Microbiology, Uppsala University, Sweden), a sensitive *L. crispatus* strain (ATCC 53545), and an AR *L. crispatus* strain (ATCC 55221). Bacterial growth was carried out in LB broth (Lennox, Sigma-Aldrich, USA) for *E. coli* and Lactobacilli MRS broth for *L. crispatus* (BD, Franklin Lakes, NJ, USA), respectively. To assess the bactericidal efficiency of the GO nanosheets, a growth inhibition curve was constructed. In detail, a 5 mg/mL stock solution of each material was dispersed in LB and Lactobacilli MRS media at 500 μg/mL. The resulting mixture was sonicated using a sonication probe (Sonics & Materials, USA) at 32 W for 15 s. The GO suspensions were diluted stepwise with the culture media to obtain a series of gradients (7.8, 15.6, 31.3, 62.5, 125, 250, and 500 μg/mL), of which 50 μL of material suspension was pipetted into 384-well polystyrene microplates. Nine replicate measurements were performed for each concentration. Using a separate plate, 50 μL of a log-phase bacterial culture (OD600 between 0.5 and 0.7) was pipetted into a 384-well plate before a plastic 384 pin replicator (Genetix Molecular Devices, USA) was used to inoculate bacteria from the plate to the plate containing the serial dilution of GO samples. Sterility and blank controls (bacterial media with no inoculation) were also included for each concentration (three replicates). After 24 h incubation at 37 °C, a SpectraMax M5 microplate reader was used to monitor OD_{600}. A growth curve was constructed using the following equation:

$$\%\ \text{growth} = \frac{A_{\text{NP,B}} - A_{\text{NP,M}}}{A_{\text{B}} - A_{\text{B,M}}} \times 100\%$$

where $A_{\text{NP,B}}$ is the absorbance of the bacterial culture in the presence of GO nanoparticles (NPs); $A_{\text{NP,M}}$ is the absorbance of the GO NPs at the respective concentrations which contain no bacteria; A_{B} is the absorbance of the bacterial culture in blank (no GO NPs) media, and $A_{\text{B,M}}$ is the absorbance of media with no bacteria.[43]

TEM Imaging of GO Interaction with Bacteria. After exposure to 125 μg/mL GO for 24 h, *E. coli* were washed with PBS (phosphate-buffered saline) and fixed with 2% glutaraldehyde in PBS. After postfixation staining in 1% osmium tetroxide in PBS for 1 h, the cells were dehydrated in a graded series of ethanol, treated with propylene oxide, and embedded in Epon. Approximately 50−70 nm thick sections were cut on a Reichert-Jung ultracut E ultramicrotome and picked up on Formvar-coated copper grids. The sections were stained with uranyl acetate and Reynolds lead citrate and examined

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on a JEOL transmission electron microscope at 80 kV in the UCLA BRI Electron Microscopy Core as previously reported.[43]

Assessment of Bacterial Membrane Lipid Peroxidation. AR *E. coli* bacteria were treated with 250 μg/mL GO samples for 6 h or 10 μM cumene hydroperoxide (positive control) for 1 h. The cells were then washed twice and incubated with 10 μM Image-iT lipid peroxidation sensor and Hoechst 33342 in culture media for 30 min. After being stained, the cells were washed three times with PBS and fluorescence readings were obtained for the reduction and oxidation of the dye at excitation/emission wavelengths of 581/591 nm (Texas Red filter set) and 488/510 nm (traditional FITC filter), respectively. The ratio of the emission fluorescence intensities at 590 to 510 nm gives a read-out for cellular lipid peroxidation. The flow cytometry analysis was carried out on a FACS Vantage SE flow cytometer from BD (Franklin Lakes, NJ). The flow cytometric data were collected and processed using Flowjo software (Ashland, OR). The confocal laser scanning microscopy analysis was performed by a TCSSP2 confocal laser scanning microscope (Leica, Wetzlar, Germany).

Assay for β-Galactosidase Release from Bacteria. Permeabilization of the *E. coli* plasma membrane was evaluated by measuring β-galactosidase activity with a Beta-Glo assay system (Promega).[44,45] In detail, 250 μL aliquots of AR *E. coli* in log phase growth (A_{600} ~ 0.7) were centrifuged to collect the cell pellets. These pellets were resuspended in 250 μL of fresh medium or each of the GO suspensions at 250 μg/mL. The supernatants were collected after 15 000 rpm centrifugation for 5 min. The positive control comprised nontreated cell pellets that were mixed with 250 μL of cell lysis buffer for the release of their β-galactosidase content. Fifty microliter aliquots of the supernatant or lysis solutions were added to the wells of 96-well plate before the addition of 50 μL of β-galactosidase substrate. Following mixing and incubation at room temperature for 2 h, the luminescence intensity was recorded in a SpectraMax M5 microplate reader. The percentage of β-galactosidase release was calculated according to the formula: $\frac{L_{GO}}{L_{ctrl} + L_{lysis}} \times 100\%$, where L_{GO} is the luminescence intensity in supernatants of GO-treated samples, and L_{ctrl} and L_{lysis} represent the luminescence values in ctrl supernatant and ctrl cell pellet lysis, respectively.

Synthesis of hGO-2 Thin Films on a Glass Substrate and on a Silicon Catheter Surface. A Laurell WS-650Sz spin-coater was used to fabricate noncovalent hGO-2 coatings on 18 mm × 18 mm × 0.15 mm glass substrates. The substrates were placed on a rotor followed by spinning at 1000 rpm. GO suspensions (2 mg/mL) were added onto the substrate surfaces using a pipet until the desired thickness was obtained. Films were dried at 50 °C overnight. These thermal treatments imparted good film−substrate adhesion as thin films were resistant to rubbing off. ImageJ was used to analyze the AFM images and determine the surface area coverage.[46]

To covalently coat a silicone catheter surface with hGO, silicone tubes were cleaned twice with acetone, DI water, and isopropyl alcohol (IPA) in a sonication bath. After 1 min plasma cleaning at high radio frequency, silicone catheters were reacted with 10 μL if APTES in 20 mL of IPA for 3 h, washed twice with IPA, DI water, and ethanol, and placed at 105 °C for 30 min. The aminated catheters were cooled to room temperature and immersed in 5 mL of 250 μg/mL carbodiimide GO solutions, which were prepared by stirring 5 mL of GO (250 μg/mL), 1-ethyl-3-(3-(dimethylamino)-propyl)carbodiimide hydrochloride (0.123 mg/mL), and *N*-hydroxysuccinimide (0.105 mg/mL) in aqueous solution for 3 h, followed by filtration and redispersion in 5 mL of DI water. After 24 h reaction with stirring, GO-modified catheters were washed twice with DI water and ethanol. The entire process was considered as a complete coating cycle. For example, to achieve a second coating, the modified catheter is once again reacted with APTES to create an aminated surface and subsequently submerged in the carbodiimide GO solution. In this study, the GO-modified catheters were coated with hGO-2 for 1, 3, 5, and 10 cycles and designated as 1C, 3C, 5C and 10C, respectively. Both the noncovalent and covalent coated GO films were further characterized by a Thermo Microscopes Autoprobe CP-Research AFM and a Renishaw inVia Reflex Raman spectroscope.

Assessment of the Bactericidal Effects of hGO-2 Films. Inhibition of bacterial growth on hGO-2-coated substrates was determined by a modified protocol.[47] AR *E. coli* in log phase were collected by centrifugation at 15 000 rpm for 5 min, washed twice with PBS, and diluted in fresh LB medium at 0.7 o.d. Bacteria were exposed to hGO-2 films by immersing the coated substrates (18 × 18 mm) or catheters (5 × 5 mm) in bacteria suspensions for 5 min. The substrates were air-dried for 30 min to allow the attachment of bacteria to the surface. The bacteria settling on catheter surfaces were then placed in a Petri dish with fresh broth and incubated at 37 °C for 1, 3, or 6 h. The growing bacteria were collected to perform 10-fold serial dilutions in 1 mL of PBS. Aliquots (0.1 mL) of each dilution were introduced to LB agar plates for 24 h incubation. Colonies were counted under a microscope, and the desired colony numbers (20−300) at appropriate dilutions were used to calculate CFU by following formula: CFU (mL^{-1}) = $N \times 10^n/0.1$, where N is the colony number on agar plates at 10^n dilution. The bacteria grown on glass substrates were stained by PI and Hoechst 33342 for 30 min. Then the glass substrates were gently rinsed three times with PBS and fixed with 70% ethanol. Bacterial cell death on the coated substrates was determined by visualizing the percentage of PI-positive cells under a Leica confocal SP2 1P/FCS microscope. In addition, SEM (Zeiss SUPRA 40VP) was used to evaluate the surface morphology of *E. coli* grown on hGO-2 films. Prior to SEM analysis, the prepared samples were mounted on the SEM stubs and coated by PELCO SC-7 sputter for 30 s at 30 mA to generate a thin Au/Pt conductive layer (2−20 nm).

Statistics. Mean and standard deviations were calculated using a minimum of three observations for each parameter, such as bacterial absorbance at 600 nm, intensity of •C on EPR spectra, % of PI-stained bacteria, *etc.* Comparisons between groups were evaluated by two-side Student's *t* test or one-way ANOVA. A statistically significant difference was assumed for $p < 0.05$.

ASSOCIATED CONTENT

Supporting Information

The Supporting Information is available free of charge on the ACS Publications website at DOI: 10.1021/acsnano.6b05692.

Antibiotics tested in resistant *E. coli*, detection of residual oxygen radicals on GO, Raman spectra of GO samples, bactericidal effects of GO in silver-resistant *E. coli*, TEM images of GO in bacterial media, durability test of carbon radicals in hGO-2, spin-coating process, Raman and AFM characterization of hGO-2 films, scheme for CFU test (PDF)

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Notes

The authors declare no competing financial interest.

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